

Project Report: IntelliSQL

Sub-title: Natural Language to SQL Generation using Gemini 3 Flash

1. Executive Summary

IntelliSQL is a Generative AI-powered application designed to bridge the gap between non-technical users and relational databases. By leveraging the **Gemini 3 Flash** model, the application allows users to query databases using plain English, eliminating the need for SQL syntax knowledge.

2. Problem Statement

Data stored in SQL databases is often inaccessible to non-technical stakeholders (HR, Marketing, Management) without the intervention of data analysts. Traditional BI tools require a learning curve. IntelliSQL solves this by providing a "Natural Language Interface for Databases" (NLIDB).

3. System Architecture

The system follows a three-tier architecture:

- User Interface (Streamlit):** A web-based frontend where users input natural language questions.
- LLM Engine (Google Gemini 3 Flash):** Processes the text input and the database schema to generate a syntactically correct SQL query.
- Database Layer (SQLite):** A local relational database that executes the generated query and returns the raw data.

4. Technical Stack

Component	Technology
Language	Python 3.x
AI Model	Google Gemini 3 Flash (via Google Generative AI API)
Web Framework	Streamlit
Database	SQLite3
Environment	Python-dotenv (for API Key security)

5. Implementation Details

A. Database Schema

For this project, a STUDENT table was implemented with the following structure:

- NAME (Varchar)

- CLASS (Varchar)
- SECTION (Varchar)
- MARKS (Int)

B. Prompt Engineering

The core "intelligence" of the app relies on a system prompt that instructs Gemini to:

- Act as an expert SQL developer.
- Understand the specific schema of the STUDENT table.
- Generate SQL code **without** markdown blocks or the word "sql" in the output.

C. Workflow

1. User enters: *"How many students are in the Data Science class?"*
2. The app combines this with the System Prompt and sends it to Gemini.
3. Gemini returns: `SELECT COUNT(*) FROM STUDENT WHERE CLASS='Data Science';`
4. The app executes this on data.db and displays the result on the UI.

6. Features & Highlights

- **Zero-Shot Learning:** The model generates queries accurately without specific training on the dataset, relying on its pre-trained reasoning.
- **Security:** Uses .env files to ensure API keys are not hardcoded.
- **Multi-page UI:** Organized into Home (Instructions), Query (Execution), and About (Project Info) for a better user experience.

7. Conclusion & Future Scope

IntelliSQL successfully demonstrates how LLMs can democratize data access.

Future Enhancements:

- Support for complex JOIN operations across multiple tables.
- Integration with cloud databases like AWS RDS or Google BigQuery.
- Data visualization (charts/graphs) for the returned SQL results.

8. References

- Google Gemini Documentation
- Streamlit API Reference
- SQLite3 Python Library Documentation